Problem Set 3 — Linear Combinations & the Sample Mean

VCE Specialist Mathematics 3&4 · Chapter 15 · Weeks 3–4

Due: start of Week 5. Answers at the back — attempt everything first.

Name: ____________________ **Due:** Monday 10 August

Method reminder: $E(aX + bY) = aE(X) + bE(Y)$; for *independent* variables $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$ (plus sign always). Combine variances, never standard deviations. $\bar{X} \sim N(\mu, \sigma^2/n)$ exactly for normal populations, approximately for large n (CLT).

Section A — Rules practice

A1. X has mean 4 and variance 9. Find the mean, variance and standard deviation of $Y = 3X - 2$. *(3 marks)*

A2. X and Y are independent with $E(X) = 5$, $\text{Var}(X) = 4$, $E(Y) = 8$, $\text{Var}(Y) = 9$. Find:

a.) $E(2X + 3Y)$

b.) $\text{Var}(2X + 3Y)$

c.) $E(X - Y)$

d.) $\text{Var}(X - Y)$

(4 marks)

A3. Explain, with variances, the difference between $2X$ and $X_1 + X_2$, where X_1, X_2 are independent copies of X. Which has the larger spread? *(3 marks)*

Section B — Discrete and normal combinations

B1. The discrete random variable X has distribution:

x	1	2	3
$P(X = x)$	0.2	0.5	0.3

a.) Find $E(X)$ and $\text{Var}(X)$. *(3 marks)* b.) The cost in dollars of producing X items is $C = 10X + 5$. Find $E(C)$ and $\mathrm{sd}(C)$. *(2 marks)*

B2. Bottles of water have volume $X \sim N(750,\ 5^2)$ mL, independently. A six-pack has total volume T.

a.) State the distribution of T. *(2 marks)*

b.) Find $P(T < 4485)$. *(2 marks)*

B3. $X \sim N(50,\ 2^2)$. Compare $P(2X > 104)$ with $P(X_1 + X_2 > 104)$, where X_1, X_2 are independent copies of X. Comment. *(4 marks)*

Section C — The sample mean and the CLT

C1. Reaction times are $N(60,\ 12^2)$ ms. A sample of 16 measurements is taken.

a.) State the distribution of $\bar{X}$. *(2 marks)*

b.) Find $P(\bar{X} < 57)$. *(2 marks)*

c.) Find $P(|\bar{X} - 60| > 6)$. *(2 marks)*

C2. A fair six-sided die has $\mu = 3.5$ and $\sigma^2 = \frac{35}{12}$. It is rolled 36 times.

a.) Which theorem lets you approximate the distribution of the mean roll $\bar{X}$? State the approximate distribution. *(2 marks)*

b.) Estimate $P(\bar{X} > 3.75)$. *(2 marks)*

C3. $X \sim \text{Bi}(400,\ 0.2)$. Use a normal approximation to estimate $P(X \geq 92)$. *(3 marks)*

Answers

A1. $E(Y) = 10$; $\text{Var}(Y) = 81$; $\text{sd}(Y) = 9$.

A2. a.) 34. b.) $16 + 81 = 97$. c.) -3. d.) $4 + 9 = 13$.

A3. $\text{Var}(2X) = 4\sigma^2$ but $\text{Var}(X_1 + X_2) = 2\sigma^2$: same mean 2μ, but $2X$ has the larger spread — doubling one draw doubles its error; two draws partly cancel.

B1. a.) $E(X) = 2.1$; $E(X^2) = 4.9$, $\text{Var}(X) = 4.9 - 4.41 = 0.49$. b.) $E(C) = 26$; $\text{sd}(C) = 10(0.7) = 7$.

B2. a.) $T \sim N(4500,\ 150)$, i.e. sd $= \sqrt{150} \approx 12.25$. b.) $z = \dfrac{4485 - 4500}{\sqrt{150}} \approx -1.22$; $P \approx 0.110$.

B3. $2X \sim N(100, 16)$: $P(2X > 104) = P(Z > 1) \approx 0.159$. $X_1 + X_2 \sim N(100, 8)$: $P = P(Z > \sqrt{2}) \approx 0.079$. The sum is less variable, so exceeding 104 is rarer.

C1. a.) $\bar{X} \sim N(60,\ 3^2)$. b.) $P(Z < -1) \approx 0.159$. c.) $2P(Z > 2) \approx 0.046$.

C2. a.) CLT ($n = 36 \geq 30$): $\bar{X} \mathrel{\dot\sim} N\big(3.5,\ \frac{35}{432}\big)$, sd ≈ 0.285. b.) $z \approx 0.88$; $P \approx 0.190$.

C3. $X \mathrel{\dot\sim} N(80,\ 64)$; $z = \dfrac{92 - 80}{8} = 1.5$; $P \approx 0.067$.